

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	XXXXXX
Project Name	AI based crop recommendation
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the code developed for the OptiCrop – AI-Based Crop Recommendation System. The project follows a well-structured coding approach with meaningful variable names, organized folders, reusable modules, and appropriate comments. These practices improve code readability, simplify maintenance, and support future enhancements such as adding more crop datasets, integrating advanced machine learning models, or including fertilizer recommendation features.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the code	Yes	Proper indentation maintained across all files.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders used for templates, static files, dataset, and model.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as N, P, K, temperature, humidity, ph, and rainfall are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like home(), predict() clearly represent their tasks.
5	Code Comments	Comments explain important sections of the code	Yes	Comments added for preprocessing, training, prediction, and routing.

6	Modular Design	Code divided into reusable modules	Yes	Separate scripts used for model training and Flask application.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Code optimized by removing unnecessary statements.
8	Error Handling	Handles invalid inputs and exceptions	Partial	Basic input validation is implemented in the prediction module.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application	Python	Handles navigation and crop prediction	High
2	Machine Learning Model	Scikit-learn	Predicts the most suitable crop	High
3	HTML Templates	HTML/CSS	Home, About, and Find Your Crop pages	High
4	Model Training Script	Python	Training and updating the ML model	High
5	CSS Stylesheet	CSS	Common styling across all webpages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure		
Readability		
Reusability		
Documentation / Comments		
Overall Score		